nd2_maxip: User Guide

Turn Nikon .nd2 files into max-projection TIFF images, one per multipoint position. No Python experience needed.

What this program does

Nikon AX confocal files (.nd2) can hold many stage positions (multipoints), each with a z-stack. This program:

- Splits the file into its individual positions.
- Collapses each z-stack into one flat image using a **maximum intensity projection** (the brightest value at each pixel across all z-slices).
- Saves the results in the format you choose with **--mode**.

You **must** pick one of five modes every time you run the program. There is no default.

Mode	What you get	Best for
simple	One 8-bit color picture per position. All channels are merged and tinted with their own colors. Standard TIFF that Windows and PowerPoint can open.	Presentations, email, figures for viewing
one-max	One grayscale intensity TIFF per position. The brightest value across all z-slices *and* all channels. Original bit depth.	Quick overview of everything in one image
multi-max	One grayscale intensity TIFF per position *per channel*. Max projection over z only. Original bit depth.	Measuring or analyzing each channel (Fiji / ImageJ)
as-nd2	One file per .nd2 holding every position and channel separately, with only the z-stack removed. Saved as an **OME-TIFF** (see note below).	Keeping the whole experiment together in one file
as-nd2-color	Same as as-nd2, but each channel also carries its display color (from the ND2 file, or from `--colors`). The pixel values stay the original intensities; only the color label is added.	Viewing the whole experiment in color in Fiji

Note on as-nd2 and as-nd2-color: no available software library can create real .nd2 files, so this mode writes an OME-TIFF (ending in `.ome.tif`) instead. It keeps the same structure: each position is a separate image series with its name, each channel keeps its name, and the pixel size is stored. Fiji/ImageJ opens it through Bio-Formats (File > Import > Bio-Formats), which lets you pick positions. With as-nd2-color, Bio-Formats shows each channel in its stored color (use the Composite view option if it opens channels separately).

One-time setup

Step 1. Install Python. Download it from python.org/downloads and run the installer. **On the first screen, tick the box "Add python.exe to PATH"** before clicking Install. This box is easy to miss and it prevents most setup problems.

Step 2. Open a command window. On Windows, press the Windows key, type `cmd`, and press Enter. (On a Mac, open the Terminal app.) This is where you will type the commands below.

Step 3. Install the required libraries. Libraries are add-on packages the program relies on. Copy this line into the command window and press Enter:

```
pip install nd2 tifffile numpy dask
```

Library	What it is used for
nd2	Reads Nikon .nd2 files
tifffile	Writes the .tif files
numpy	Does the image math (the projection)
dask	Lets large files load piece by piece instead of all at once

You only need to do this once per computer. It may take a minute and print a lot of text; that is normal.

Step 4. Save the program. Put `nd2_maxip.py` in any folder you like, for example `C:\nd2tools`.

Running the program

1. Go to the folder with the program. In the command window, type `cd` followed by a space, then drag the folder from File Explorer into the window (this fills in the path for you) and press Enter:

```
cd C:\nd2tools
```

2. Run it on a file. Use quotes around any path that contains spaces:

```
python nd2_maxip.py "C:\data\experiment1.nd2" --mode simple
```

Replace `simple` with the mode you want (simple, one-max, multi-max, as-nd2 or as-nd2-color). If you leave out `--mode`, the program stops and tells you it is required.

If `python` is not recognized, try `py` instead, or see Troubleshooting below.

Files are saved in the same folder as the .nd2 file (or in the folder you give with `-o`). Every file name follows this pattern:

```
<index>_<nd2 name>_P<position #>_<position name>_C<channel #>_<channel name>
```

Part	Meaning
index	A running count of every file the program saves (0001, 0002, ...), across all .nd2 files in that run. It keeps names from ever overlapping and shows how far along the run is.
P<position #>_<position name>	The multipoint number and the name set in NIS-Elements. If a position has no name, only the number is used.
C<channel #>_<channel name>	Only in multi-max mode (one file per channel).

Examples:

```
multi-max:    0003_exp1_P002_Control_C1_DAPI_max.tif
one-max:      0002_exp1_P002_Control_maxall.tif
simple:       0002_exp1_P002_Control_rgb.tif
as-nd2:       0001_exp1_maxip.ome.tif
as-nd2-color: 0001_exp1_maxip_color.ome.tif
```

Characters Windows does not allow in file names (such as / : ?) are replaced with underscores, and spaces in names become underscores. If the file has a time series, simple mode saves one image per time point (adds `_T001`, `_T002`, ...) and multi-max / one-max save the time points as a stack inside each file.

Common examples

I want to...	Type this
Color images for a presentation	`python nd2_maxip.py "C:\data\a.nd2" --mode simple`
Separate intensity image for every channel	`python nd2_maxip.py "C:\data\a.nd2" --mode multi-max`
Process every .nd2 in a folder	`python nd2_maxip.py "C:\data\run1" --mode one-max`
Save results to a different folder	`python nd2_maxip.py "C:\data\a.nd2" --mode as-nd2 -o "C:\data\output"`
Replace files that already exist	`python nd2_maxip.py "C:\data\a.nd2" --mode multi-max --overwrite`

All options

Option	What it does
`-o FOLDER`	Save the output in FOLDER instead of next to the .nd2 file. The folder is created if needed.
`--overwrite`	Replace files that already exist. Without this, existing files are skipped.

Option	What it does
`--mode MODE`	**Required.** One of simple, one-max, multi-max, as-nd2, as-nd2-color (see the table at the top).
`--colors blue,red`	Simple and as-nd2-color modes: choose the color of each channel, in channel order. Names: red, green, blue, cyan, magenta, yellow, orange, gray. Hex codes like `#0000ff` also work. Normally not needed, because colors are read from the file.
`--low 0.5` `--high 99.8`	Simple mode only. **Optional; off by default.** By default, brightness is a fixed scale (0 up to the camera's maximum), so background noise stays dark but dim images stay dim. Giving --low and/or --high turns on auto-contrast, which stretches each channel between those two percentiles (the one you leave out uses 0.5 or 99.8). Lower --high to brighten. See the note below.

About auto-contrast: it stretches each channel separately, so a channel with almost no real signal (just background noise) gets amplified until the noise looks like a colored haze over the whole image. That is why it is off by default. If you turn it on and see this, raise --low (for example 50) so the background falls to black, or leave auto-contrast off.

In simple and as-nd2-color modes the program prints the channel colors it found, for example `channel colors from file (RGB): [(0, 0, 255), (255, 0, 0)]`. That is blue then red. Check that this matches what you expect.

Good to know

- **Use multi-max (or as-nd2) for measurements.** Simple images are adjusted for display, so their brightness values are no longer the original data. one-max keeps the original values but merges channels, so it is not suited to per-channel measurements.
- The original .nd2 file is never changed.
- Opening the intensity TIFFs in Fiji: use File > Open.

Troubleshooting

Problem	What to try
`'python' is not recognized`	Try `py` in place of `python`. If that fails, reinstall Python and tick "Add python.exe to PATH".
`No module named 'nd2'` (or tifffile, numpy, dask)	The libraries are not installed. Run the `pip install` line from Step 3 again. If `pip` is not recognized, use `python -m pip install nd2 tifffile numpy dask`.
`error: the following arguments are required: --mode`	Add `--mode simple` (or one-max, multi-max, as-nd2, as-nd2-color) to the end of the command.
`No .nd2 files found.`	The path is wrong or missing quotes. Check the spelling and put quotes around the path.

Problem	What to try
Skipped: "exists; use --overwrite"	A file with that name already exists. Add `--overwrite` or choose a new folder with `-o`.
Warning: "could not read channel colors"	The file's colors could not be read, so default colors were used (also use this if the colors simply look wrong). Set them yourself with `--colors`, for example `--colors blue,red`.
Colored haze or "static" over the whole image (simple mode)	Auto-contrast is amplifying a channel that has almost no real signal. Run again **without** `--low` / `--high` (the default), or raise `--low` (for example 50) so the background goes black.
Simple-mode image looks too dark	The default fixed scale shows the data as recorded, so dim images stay dim. Add `--high 99.9` (or a lower number for a brighter picture) to turn on auto-contrast.
Out-of-memory error or very slow	Close other programs and process one file at a time.